- **Circumduction**: The muscular and joint movement that entails complete 360° movement
- **Inversion:** The turning of a joint inward
- **Eversion:** The turning of a joint outward
- **Plantar flexion:** Movement of the foot downward
- **Dorsiflexion:** Movement of the foot upward

## Cartilage

Cartilage is an important structural component of the body. It is less hard than bones and more flexible than bones and helps in articulations It protects the ends of the bone. It is found in elbows , knees and ankle.




Cartilage does not have a blood supply, nerves and marrow.

## Tendons

A tendon is a tough band of fibrous connective tissue that usually connects muscle to bone and is capable of withstanding tension. It is made up of collagen.

## Ligaments

A ligament is a short band of tough, flexible fibrous tissues, tissues that connects bones to bone to form a joint. It also maintain position of the organs.

## Joints

Joint or articulation is the connection made between bones in the body which link the skeletal system into a functional whole.

## Types of Moveable Joints

- **Hinge joint,** e.g. interphalangeal joints – fingers and toes.
- **Ball and socket joint**, e.g. **hip** and **shoulder** joints
- **Pivot joint**, e.g. atlantoaxial joint between the atlas and axis – **neck**
- **Gliding joint ("Condyloid" joint)**, e.g. between radius, scaphoid and lunate bones – wrist.

## Structural and functional

**Structural (binding tissue)**

- **Fibrous joint** - joint by connective tissue.
- **Cartilage joint** - joint by cartilage
- **Synovial joint** - not directly joined.
- **Facet joint** - between two vertebrae.

**Functional (movement)**

- **Synarthrosis** means no mobility (skull)
- **Amphiarthrosis** means slight mobility (vertebrae)
- **Diarthrosis** means freely movable (knee)

## Fascia

A fascia is a band or sheet of connective tissue, primarily collagen, beneath the skin that attaches, stabilize, enclose and separates muscles and other internal organs.

## It is classified as

- Superficial fascia
- Deep fascial fascia
- Visceral fascia
- Parietal fascia.

## Diseases related to the bone

- Arthritis
- Rheumatoid arthritis
- Osteoporosis
- Osteoarthritis
- Osteomyelitis
- Fractures

## Functions of musculo skeletal system

The human skeleton and the skeletal system perform several functions.

1. The protection of the vital organs of the body



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